

Depression in numbers: A comparison of 195 countries and of men and women in 2017

Emma Schwarz (6402721)

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Introduction

As depressive disorders have become more prevalent over the past years, with an estimate of 5.2 % of adults affected globally [1], a dataset of "Global trends in mental health disorders" was used. One might think that depression differs a lot between countries and is more common in women than in men. In this project both statements were analyzed, using numbers. The public data used included the share of people with depressive disorders per country, as well as the share of men and women suffering from them. The research questions answered in this report are: **(1) In which country was depression most common in 2017? (2) Was depression more common in women or men in 2017?**

Data sample

The dataset [2] is a public file, called `Mental health Depression disorder Data.csv` from Kaggle. Each row in the dataset included one country in one year and the share of its population (in %) with a mental health disorder. The downloaded file had 108 533 rows and showed 4 tables stacked on top of each other, from which two tables were used. The two tables analyzed gave the share of the population with depression (`Depression (%)`) and the share of men and women affected (`Prevalence in males (%)` and `Prevalence in females (%)`) for every country.

Methods

The data was cleaned using scripts, removing rows without depression value (in table two 41 339 rows only contain a population). Additionally, groups of countries and regions e.g. "Western Europe", were removed to avoid counting people twice. Furthermore, rows with the entry "World", adding all the countries together were removed (28 rows, one per year) ending up with 5 460 rows per table, showing data from 195 countries of 28 years from 1990 to 2017. Finally, only focusing on the most recent year (2017) shown in the data, the last cleaning step was to remove the data from before 2017, giving one value per country, meaning 195 values. For the analyses of the data R was used, using one script for each question and the plots were generated with Python, using pandas and matplotlib. The sub-sampling technique was applied, splitting the 195 countries into 5 groups of 39. For each subsample the mean was calculated, using $\mu = \sum x_i/N$ (with $N = 39$) from which the mean of the 5 groups means was the final result needed with the sample standard deviation as the uncertainty, using $s = \sqrt{\sum(x_i - \bar{x})^2/(N-1)}$ with $N = 5$. The uncertainty for a single country could not be given since it was only one value in the file.

Results and discussion

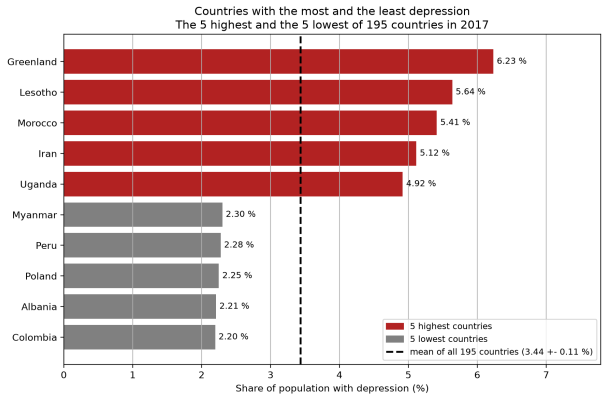


Figure 1: The five countries with the highest and the five countries with the lowest share of people with depression in 2017, with the dashed line representing the mean of all countries

Question 1. Figure 1 shows the ranking from both ends. It is clearly visible that depression was most common in Greenland (6.23 %), followed by Lesotho (5.64 %), Morocco (5.41 %), Iran (5.12 %) and Uganda (4.92 %). It was least common in Colombia (2.20 %), closely followed by Albania (2.21 %) and Poland (2.25 %). Greenland has about 2.8 times the share of Colombia. Looking at the average across all countries, 3.44 ± 0.11 % of the population suffered from depression and the countries differed from each other with a standard deviation of 0.65 %. From the 195 countries most (140) of them lie close to the mean between 2.79 and 4.08 % (mean ± one standard deviation), with Greenland as a clear outlier with 4.3 standard deviations above the mean.

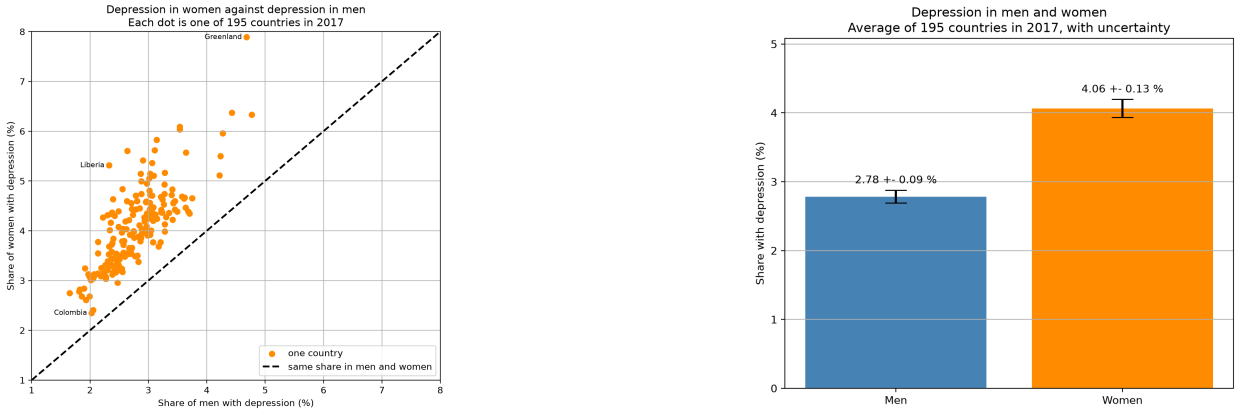


Figure 2: Left: shows the share of women against the share of men suffering from depression in 2017, with each dot representing a country and the dashed line marking where women and men would be equally affected. Right: shows the average share of women and the average share of men suffering from depression across the 195 countries, with the error bars showing the uncertainty.

Question 2. In Figure 2 (left) every dot representing a country is above the dashed line, showing that depression was more common in women in 2017 in all 195 countries. Looking at the averages, 2.78 ± 0.09 % of men and 4.06 ± 0.13 % of women were suffering from depression (Figure 2, right). The difference was also calculated to compare women and men (women − men) as well as the ratio (women ÷ men) for every country first and then using the sub-sampling technique in the same way as before. The difference calculated is 1.28 ± 0.07 percentage points, being almost 20 times its uncertainty ($1.281/0.065 = 19.7$), clearly showing us that the difference is not due to random chance. The mean ratio calculated is 1.47 ± 0.03: with the share of women suffering from depression 47% higher than that of the men. The ratio in Liberia was largest (with: men 2.32 %, women 5.31 %, ratio 2.29) and Colombia the smallest (with: men 2.03 %, women 2.35 %, ratio 1.16).

Conclusion. To conclude it is possible to say that in 2017 depression was most common in Greenland, with more women than men affected in all 195 countries. The analysis had limitations, with the data showing that the differences exist but not why they exist. Additionally, it analyses only one year and only 5 groups were used in the sub-sampling technique, making it a rough estimate. Finally, the uncertainty for every country could not be calculated, meaning the resulting order with almost the same numbers (e.g. Albania and Colombia) should not be interpreted too strictly.

References

[1] World Health Organization. *Depressive disorder (depression)*. Fact sheet. Available at: `https://www.who.int/news-room/fact-sheets/detail/depression`.

[2] The Devastator. *Global Trends in Mental Health Disorder*. Kaggle. Available at: `https://www.kaggle.com/datasets/thedevastator/uncover-global-trends-in-mental-health-disorder`.